

YAROL'IN, Russian Federation

WMO: 242190

Lat: 67.133N

Lon: 108.533E

Elev: 774

StdP: 14.29

Time Zone: 9.00 (E09)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-65.5	-61.9	-67.8	0.1	-64.9	-64.4	0.1	-61.9	20.9	7.9	17.6	6.4	0.3	260	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	82.2	64.3	77.9	62.2	73.5	59.5	65.7	78.9	63.1	75.3	60.7	71.4	5.8	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.0	82.6	70.6	58.2	74.5	67.8	55.8	68.1	65.3	31.1	79.0	29.0	75.2	27.3	71.6	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.3	15.6	13.6	DB	-66.8	88.0	3.7	4.5	-69.5	91.3	-71.6	93.9	-73.7	96.4	-76.4	99.7	(4)
(5)			WB	-66.8	69.0	3.7	4.3	-69.5	72.1	-71.6	74.6	-73.7	77.0	-76.4	80.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.0	-36.3	-25.3	-3.8	13.9	33.5	54.8	58.1	52.5	38.5	15.9	-18.8	-29.6	
	DBStd	36.26	18.75	18.76	17.15	12.92	10.42	10.22	7.46	7.04	8.46	13.92	16.24	18.79	
	HDD50	14132	2677	2108	1667	1083	519	64	14	54	358	1057	2063	2468	
	HDD65	19040	3142	2528	2132	1533	978	332	238	393	796	1522	2513	2933	
	CDD50	622	0	0	0	0	6	209	266	130	11	0	0	0	
	CDD65	56	0	0	0	0	0	27	25	4	0	0	0	0	
	CDH74	860	0	0	0	0	0	410	363	80	7	0	0	0	
	CDH80	207	0	0	0	0	0	120	76	11	0	0	0	0	
Wind	WSAvg	4.8	3.4	3.6	5.2	6.4	5.9	5.8	4.6	5.0	5.3	5.1	3.7	3.7	
Precipitation	PrecAvg	15.3	0.6	0.4	0.6	0.9	1.3	2.2	2.1	2.5	1.6	1.5	0.9	0.7	
	PrecMax	21.0	1.0	1.0	1.0	1.7	2.3	3.5	3.3	4.9	2.9	2.1	1.6	1.8	
	PrecMin	11.2	0.2	0.1	0.2	0.3	0.6	0.6	0.6	0.9	0.4	0.7	0.4	0.2	
	PrecStd	2.0	0.2	0.2	0.2	0.3	0.5	0.7	0.7	0.9	0.6	0.4	0.3	0.3	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.0	20.6	39.5	45.7	67.5	88.2	85.6	81.5	74.3	45.0	23.1	22.6	
		MCWB	17.5	18.5	34.7	37.0	52.6	66.2	64.9	63.3	58.9	40.3	21.6	20.2	
	2%	DB	11.2	14.6	34.4	40.9	60.9	83.6	82.0	75.3	61.7	40.5	16.2	14.8	
		MCWB	9.9	12.9	30.1	34.5	49.8	63.6	65.5	60.3	51.1	36.7	14.6	12.9	
	5%	DB	3.2	10.9	29.1	37.8	54.6	78.9	77.9	70.7	55.8	36.9	10.8	9.3	
		MCWB	2.4	9.4	26.1	33.2	45.2	61.0	63.2	58.2	47.8	33.9	9.8	8.0	
	10%	DB	-4.7	4.0	23.7	34.0	49.5	73.3	74.0	65.9	51.3	33.2	5.5	1.1	
		MCWB	-5.4	3.3	21.1	30.0	42.1	59.1	60.0	55.9	44.7	30.7	4.7	0.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.4	18.9	34.8	38.6	53.3	68.3	70.5	65.4	58.9	40.3	21.1	20.1
			MCDB	19.3	20.7	38.4	43.8	65.8	83.9	80.2	77.4	73.8	43.9	22.2	22.6
2%		WB	9.8	13.1	30.1	35.5	49.9	65.1	67.2	62.4	52.0	37.1	14.6	13.1	
		MCDB	11.0	14.6	34.1	40.8	59.7	79.7	78.6	72.9	59.8	40.1	15.9	14.5	
5%		WB	2.2	9.5	26.0	33.2	46.0	62.6	63.9	59.8	48.5	34.1	9.8	7.9	
		MCDB	2.9	11.0	29.0	37.5	53.0	75.8	75.8	68.0	54.5	36.9	10.7	9.3	
10%		WB	-5.2	3.1	21.0	30.2	42.2	60.0	61.3	57.2	45.6	30.6	4.8	0.3	
		MCDB	-4.6	3.9	23.4	33.8	49.2	72.3	71.8	63.9	50.9	33.0	5.4	1.1	
Mean Daily Temperature Range		5% DB	MDBR	15.2	20.0	26.4	24.1	16.3	19.3	20.4	18.4	14.6	14.0	16.3	16.2
			MCDBR	20.9	19.8	21.8	23.8	23.1	28.1	28.3	29.3	24.2	11.5	21.0	23.8
	MCWBR		19.8	19.2	19.1	19.4	14.1	12.5	14.3	16.0	15.0	8.7	19.7	22.6	
	5% WB	MCDBR	20.4	20.2	22.0	22.3	22.1	25.5	24.9	23.7	21.8	11.5	21.2	23.9	
		MCWBR	19.8	19.4	19.4	18.6	14.2	12.6	13.5	13.9	14.6	9.2	20.3	22.7	
Clear-Sky Solar Irradiance	taub	0.193	0.240	0.291	0.320	0.348	0.364	0.430	0.413	0.313	0.249	0.182	N/A		
	taud	1.860	2.024	2.047	2.073	2.195	2.407	2.195	2.194	2.455	2.164	1.886	N/A		
	Ebn at Noon	47	183	234	261	267	267	239	228	237	180	46	0		
	Edh at Noon	4	17	29	38	37	32	38	34	20	14	3	0		
All-Sky Solar Radiation	RadAvg	19	225	754	1435	1806	1742	1564	1047	588	247	49	0		
	RadStd	2	12	43	81	114	161	135	135	64	28	4	0		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page